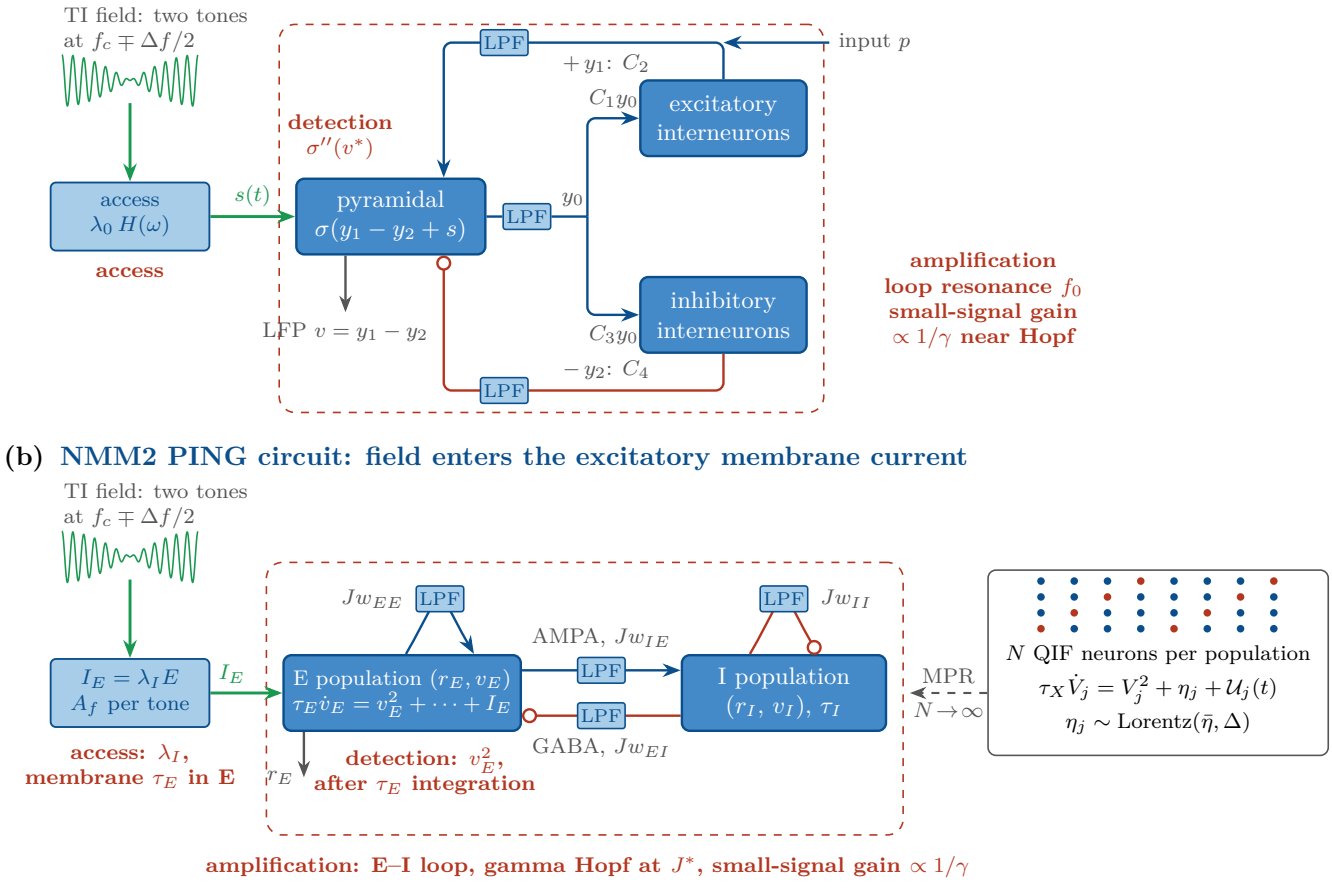

(a) Jansen–Rit column: field enters the pyramidal sigmoid
TI field: two tones
at $f_c \mp \Delta f/2$
access
$\lambda_0\, H(\omega)$
access
$s(t)$
detection
$\sigma''(v^*)$
pyramidal
$\sigma(y_1 - y_2 + s)$
LPF
LPF
LPF
input p
$+y_1$: C_2
$C_1 y_0$
excitatory
interneurons
y_0
inhibitory
interneurons
$C_3 y_0$
$-y_2$: C_4
LFP $v = y_1 - y_2$
amplification
loop resonance f_0
small-signal gain
$\propto 1/\gamma$ near Hopf
(b) NMM2 PING circuit: field enters the excitatory membrane current
TI field: two tones
at $f_c \mp \Delta f/2$
$I_E = \lambda_I E$
A_f per tone
I_E
access: λ_I,
membrane τ_E in E
Jw_{EE}
LPF
E population (r_E, v_E)
$\tau_E \dot{v}_E = v_E^2 + \cdots + I_E$
AMPA, Jw_{IE}
LPF
LPF
GABA, Jw_{EI}
r_E
detection: v_E^2,
after τ_E integration
LPF
Jw_{II}
I population
$(r_I,\, v_I)$, τ_I
MPR
$N \to \infty$
N QIF neurons per population
$\tau_X \dot{V}_j = V_j^2 + \eta_j + \mathcal{U}_j(t)$
$\eta_j \sim$ Lorentz$(\bar{\eta}, \Delta)$
amplification: E–I loop, gamma Hopf at J^*, small-signal gain $\propto 1/\gamma$